ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

DISTANCE LEARNING PROGRAMME
(Academic Session : 2024 - 2025)

JEE (Advanced)
TEST # 14
29-12-2024

JEE(Main + Advanced) : LEADER TEST SERIES / JOINT PACKAGE COURSE

Test Type : Review (Unit Test # 06, 07, 08, 09 & 10)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	C	A	A		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,C,D	A,C,D	A,B,C,D			
SECTION-II (i)	Q.	1	2	3	4		
	A.	6.25	1.25	7.00	0.75		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	3	2	3	9	3	5

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	D	C	C		
SECTION-I (ii)	Q.	5	6	7			
	A.	C,D	A,D	A,B,C			
SECTION-II (i)	Q.	1	2	3	4		
	A.	5.00	8.00	7.40	0.02		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	9	6	8	5	6	3

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	B	B	B		
SECTION-I (ii)	Q.	5	6	7			
	A.	A,B,D	A,B,C	A,B,C,D			
SECTION-II (i)	Q.	1	2	3	4		
	A.	156.50	1.00	64.00	3.00		
SECTION-II (ii)	Q.	5	6	7	8	9	10
	A.	75	26	2	2	1	81

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

2. Ans (C)

$$K = \frac{1240}{620} - 1.5 = 0.5 \text{ eV} = eEd$$

$$= e \times \frac{2\epsilon_0}{\epsilon_0} d$$

$$d = \frac{1}{4} m$$

3. Ans (A)

From screw guage

zero error = + 0.20 mm

Dia of Chalk = 3.5 – 0.2

= 3.3 mm

for vernier calliper

zero error = – 0.05 cm

Thus it must show = 0.33 – 0.05

= 0.28 cm

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4. **Ans (A)**

$\Delta\phi = \frac{2\pi}{\lambda}\Delta x$ & $\Delta x = \left(\frac{y}{D}\right) \times d$

$y_1 = \frac{3\lambda D_1}{d}$

Now for y_2

$I_{max}\cos^2\frac{\Delta\phi}{2} = I_{max} \times \frac{3}{4} \Rightarrow \Delta\phi = \frac{\pi}{3} + 2n\pi$

& this can be $4\pi + \frac{\pi}{3}$ or $6\pi - \frac{\pi}{3}$ $6\pi - \frac{\pi}{3}$

$\Rightarrow \left(\frac{y_2}{D_2} \times d\right)\frac{2\pi}{\lambda} = \Delta\phi \Rightarrow y_2 = \frac{17D_2\lambda}{6d}, \frac{13D_2\lambda}{6d}$

$\because\ y_1 = y_2 \Rightarrow \frac{13D_2}{6} = 3D_1$

finally & $D_2 = \frac{18}{13}D$

$v = \frac{D_2 - D_1}{\Delta t} = \frac{5D}{13}$

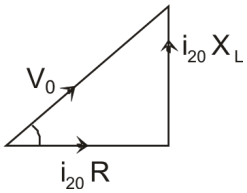
PART-1 : PHYSICS

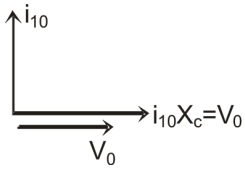
SECTION-I (ii)

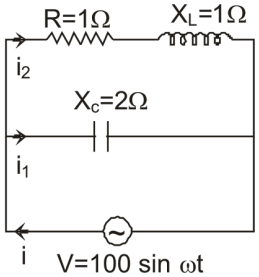
5. **Ans (A,C,D)**

$i_{20} = \frac{V_0}{\sqrt{R^2 + X_2^2}} = 50\sqrt{2}\,A$

$\Rightarrow i_{10} = 50\,A$





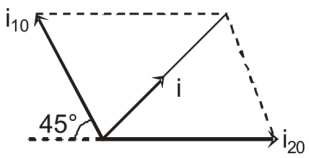


Phase difference between v and i is $\theta = 0°$

$i = i_1 + i_2$

$i_0 = 50$

$P_{av} = \frac{i_0 v_0}{2}\cos\theta = 2500\,W$



$z = \frac{V_0}{i_0} = 2\Omega.$

6. **Ans (A,C,D)**

$13.6z^2\left[\frac{1}{1^2} - \frac{1}{(2n)^2}\right] = 204$

$13.6z^2\left[\frac{1}{n^2} - \frac{1}{(2n)^2}\right] = 40.8$

solving, $n = 2$, $z = 4$

7. **Ans (A,B,C,D)**

In standing wave energy is confined between node & antinode.

Node and antinode do not transfer its energy but element between node & antinode transfers its energy to other element.

PART-1 : PHYSICS

SECTION-II (i)

1. **Ans (6.25)**

$-mv \cdot \frac{dv}{dl} = \mu(mg + qE_0)$

$-m\int_{v_0}^{0} vdv = \mu(mg + qE_0)\int_0^l dl$

$l = \frac{mv_0^2}{2\mu(mg + qE_0)}$

$l = \frac{1(100)}{2 \times 4(10 + 2 \times 5)} = 6.25\ m$

2. **Ans (1.25)**

$N = mg + qE_0$

and $-m\frac{dv}{dt} = \mu N$

$-m\frac{dv}{dt} = \mu(mg + qE_0)$

or $-m\int_{v_0}^{0} dv = \mu(mg + qE_0)\int_0^t dt$

Thus $t = \frac{mv_0}{\mu(mg + qE_0)}$

$\frac{1(10)}{0.4(1 \times 10 + 2 \times 5)} = 1.25\ s$

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3. **Ans (7.00)**

$$= \left(\frac{1}{2}mR^2\right) w_1 - \left(\frac{1}{2}m\left(\frac{R}{2}\right)^2\right) w_2$$

$$= \left[\frac{1}{2}mR^2 + \frac{1}{2}m\left(\frac{R}{2}\right)^2 + m\left(\frac{R}{2}\right)^2\right] w_f$$

$$= \frac{w_1}{2} - \frac{w_2}{8}$$

$$= \left(\frac{1}{2} + \frac{1}{8} + \frac{1}{4}\right) \times \omega_f$$

$$= \frac{1}{2} - \frac{3}{8} = \frac{4+1+2}{8} w_f$$

$$\beta = w_f = \frac{1}{7}$$

$$49\beta = 49 \times \frac{1}{7} = 7$$

4. **Ans (0.75)**

$$\Delta L = L_F - L_i$$

$$= \left\{\frac{1}{2}m\left(\frac{R}{2}\right)^2 w_2 + \frac{1}{8}w_1\right\}$$

$$= \left\{\frac{1}{2}m\left(\frac{R}{2}\right)^2 + m\left(\frac{R}{2}\right)^2\right\} W_2$$

$$- \left\{-\frac{1}{2}m\left(\frac{R}{2}\right)^2\right\} w_1$$

$$= mR^2 \left\{\frac{3}{8} + \frac{3}{8}\right\}$$

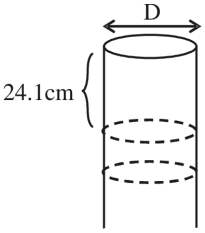
$$= 10 \times 1 \times \frac{3}{4}$$

$$\tau = \frac{\Delta L}{\Delta t} = 0.75$$

PART-1 : PHYSICS

SECTION-II (ii)

5. **Ans (3)**



we have, end correction in a resonance tube is

$$e = \frac{L_2 - 3\ell_1}{2}$$

$\ell_1 = 24$ km, $\ell_2 = 79.1$ cm, $e = 0.3D$

$$\Rightarrow 0.3D = \frac{79.1 - 3 \times 24.1}{2}$$

$$0.3D = \frac{1.8}{2} = 0.9$$

$D = 3$ cm

6. **Ans (2)**

Excess pressure in a bubble of gas in a liquids is

Pressure out side the bubble is $P_0 + rgh$

When P_0 is atmospheric pressure $\frac{2s}{r}$

Therefore pressure inside the bubble is $P_0 + \rho gh$

$$P_0 + \rho gh + \frac{2s}{R}$$

$$1.01 \times 10^5 \times 10^3 \times 10 \times \frac{10}{100} + \frac{2 \times 8 \times 10^{-2}}{0.02 \times 10^{-3}}$$

$$1.01 \times 10^5 + 10^3 + \frac{16 \times 10^{-2}}{2 \times 10^{-3}}$$

$$1.01 \times 10^5 + 10^3 + 0.08 \times 10^5$$

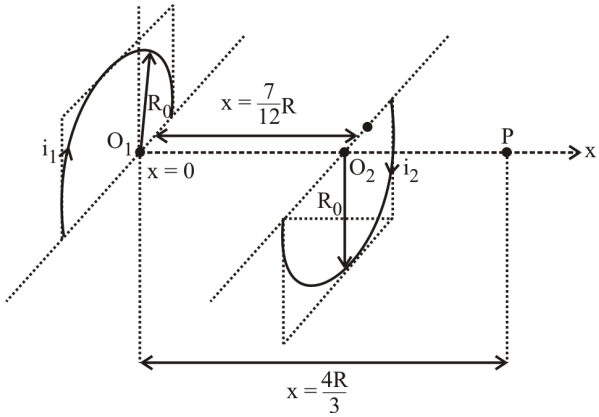
$$1.01 \times 10^5 + 0.01 \times 105 + 0.08 \times 10^5$$

$$= 105\ (1.01 + 0.01 + 0.08)$$

$$= 1.1 \times 10^5 = 110 \times 10^3 = 55x \times 10^3$$

So $x = 2$

7. **Ans (3)**



$$i_1 \sin^2 37 \cos 37 = i_2 \sin^2 53 \cos 53$$

$$i_2 = \frac{3}{4} i_1$$

$$\frac{\mu_0 i_1}{4R} \sin^3 37° + \frac{\mu_0}{4R} \times \frac{3}{4} i_1 \sin^3 53°$$

$$\frac{\mu_0 i_1}{4R}\left[\frac{27}{125} + \frac{3}{4} \times \frac{64}{125}\right]$$

$$\frac{\mu_0 i_1}{4R} \times \frac{75}{125} = 3\frac{\mu_0 i_1}{20R}$$

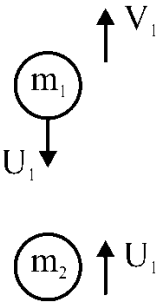
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8. **Ans (9)**

$$V_1 = \frac{m_1 - m_2}{m_1 + m_2}U_1 - \frac{2m_2}{m_1 + m_2}U$$

$$V_1 = \frac{m_1 - m_2 - 2m_2}{m_1 + m_2}U$$

$$V_1 = \left(\frac{m_1 - 3m_2}{m_1 m_2}\right)U$$



V_1 will be maximum when $m_2 >>> m_1$

$\therefore \quad V_1 = -3U$

$\therefore \quad h = 9h$

9. **Ans (3)**

$$\sigma = -\left(\frac{\frac{\Delta d}{d}}{\frac{\Delta h}{h}}\right)$$

$$v_w = 2A_i = A_f(2 + 0.08 \times 10^{-2})$$

$$\Rightarrow \frac{A_f}{A_i} = \frac{2}{2 + 0.08 \times 10^{-2}}$$

$$\frac{A_f - A_i}{A_i} = \frac{-0.08 \times 10^{-2}}{2 + 0.08 \times 10^{-2}} \approx \frac{-0.08 \times 10^{-2}}{2}$$

$$\therefore \frac{\Delta A}{A} = -\frac{0.08 \times 10^{-2}}{2} = 2\frac{\Delta d}{d}$$

$$\Rightarrow \frac{\Delta d}{d} = -0.02 \times 10^{-2}$$

$$\therefore \sigma = \frac{0.02 \times 10^{-2}}{\frac{0.12 \times 10^{-2}}{2}} = \frac{1}{3}$$

$$\Rightarrow \frac{1}{\sigma} = 3$$

10. **Ans (5)**

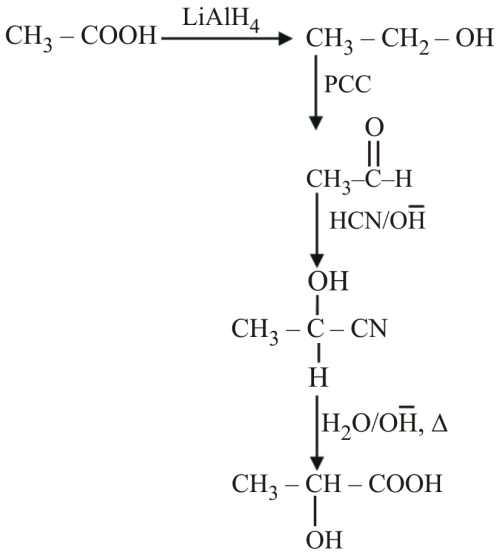
$$\frac{1}{2}\frac{m \times 4m}{m + 4m}(v - 0)^2 \geqslant W_0$$

$$\Rightarrow \frac{1}{2}4mv^2 \geqslant 5w_0$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (D)**



2. **Ans (D)**

(1) NH_3 has higherboiling point than AsH_3 and PH_3 on account H-bonding. SbH_3 has highest boiling point on account of highest molecular weight.

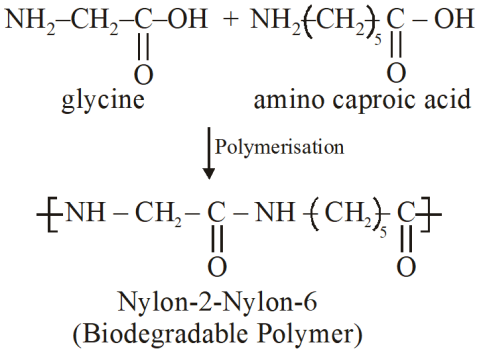
(2) As the size of element increases, the strength of M–H bond decreases and thus stability decreases.

(3) Basicity decreases down the group.

3. **Ans (C)**

$$Z = \frac{1}{\sqrt{2}}\pi\sigma^2\sqrt{\frac{8RT}{\pi M}}.\frac{P}{KT}$$

4. **Ans (C)**

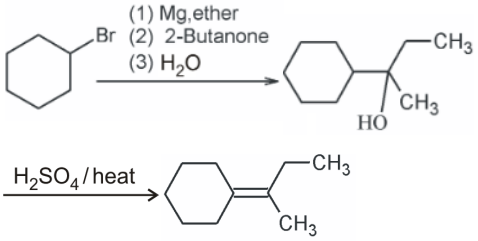


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PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (C,D)



6. Ans (A,D)

Freshly prepared

$FeSO_4$ solution + NO_3^- solution + conc. H_2SO_4 $\longrightarrow$ $[Fe(H_2O)_5(NO)]^{+}(SO_4)$ Brown ring complex or Iron nitrosyl complex

Fe^+ : $[Ar]3d^7$

Unpaired electron = 3

Paramagnetic

7. Ans (A,B,C)

$x(s) \longrightarrow x^{+2}(0.001\ M) + 2e^-$ (anode)

$y^{+2}(0.1\ M) + 2e^- \longrightarrow y(s)$ (cathode)

$$E_{cell} = E^\circ_{cell} - \frac{0.06}{2}\log\frac{x^{+2}}{y^{+2}}$$

$E_{cell} = E^\circ_{cell} + 0.06$

(A) Cd and Ni $E^\circ_{cell} = +0.4 - 0.24$; $E_{cell} = 0.22$

(B) Cd and Fe $E^\circ_{cell} = -0.04$; $E_{cell} = 0.02$

(C) Ni and Pb $E^\circ_{cell} = 0.11$; $E_{cell} = 0.17$

(D) Ni and Fe $E^\circ_{cell} = -0.2$; $E_{cell} = -0.14$

since in (A) (B) (C) E_{cell} is positive hence answer is (A) (B) (C).

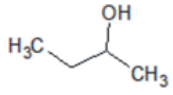
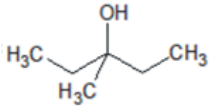
PART-2 : CHEMISTRY

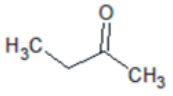
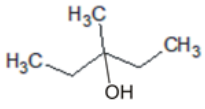
SECTION-II (i)

1. Ans (5.00)

(P) CH_3COOH (Q) CH_3CH_2OH

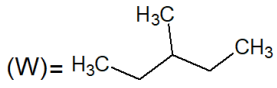
(R) CH_3CH_2Br (S) CH_3CHO

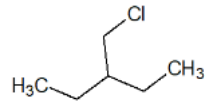
(T)  (U) 

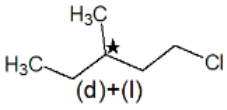
(V)  (W) 

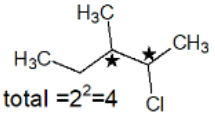
(Q),(R),(S),(T),(V) can give yellow ppt when react with NaOI

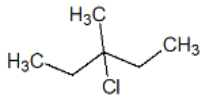
2. Ans (8.00)

(W)= 

Monochloro products = 







3. Ans (7.40)

$Na_2CO_3 + HCl \longrightarrow NaHCO_3 + HCl$

0.1M

$[H^+] = \sqrt{K_{a_1}K_{a_2}} = \sqrt{16 \times 10^{-16}} = 4 \times 10^{-8}$

$pH = 8 - \log 4 = 8 - 0.6 = 7.4$

4. Ans (0.02)

$HCO_3^- \rightleftharpoons H^+ + CO_3^{2-}$, $K_{a_2} = 8 \times 10^{-10}$

0.1 M 4×10^{-8} M

$$8 \times 10^{-10} = \frac{(4 \times 10^{-8})\ [CO_3^{2-}]}{0.1}$$

$\Rightarrow [CO_3^{2-}] = 2 \times 10^{-3}$ M

PART-2 : CHEMISTRY

SECTION-II (ii)

5. Ans (9)

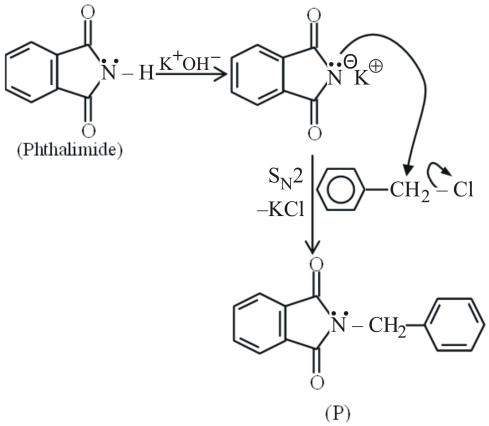
n = 3

$$\text{P.E.} = -2 \times \text{T.E.} = \frac{-2 \times 13.6 \times (1)^2}{(3)^2}$$

6. Ans (6)

Be and Mg compounds do not show flame colouration test.

7. Ans (8)



Total number of π-bonds present in product P is 8

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8. **Ans (5)**

i, iii, v, vi, viii

9. **Ans (6)**

$\log \frac{x}{m} = \log k + \frac{1}{n}\log p$

$\frac{1}{n} = \tan 45° = 1$

$\log k = 0.3010$

$\Rightarrow k = 2$

then $\frac{x}{m} = KP^{\frac{1}{n}} = 2 \times (3)^1 = 6$

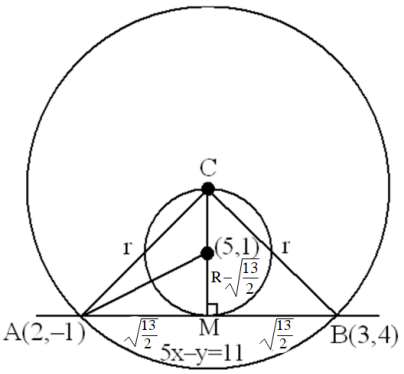
10. **Ans (3)**

NH_4NO_2, $(NH_4)_2Cr_2O_7$, NH_4MnO_4

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B)**



$AB = \sqrt{26}$

$r^2 = CM^2 + AM^2$

$= \left(2 \times \sqrt{\frac{13}{2}}\right)^2 + \left(\sqrt{\frac{13}{2}}\right)^2$

$r^2 = \frac{65}{2}$

2. **Ans (B)**

$x^2 + y^2 + 2gx + 2fy + c = 0$

It touches the x-axis at origin

$\Rightarrow \quad c = 0$

$\Rightarrow \quad \sqrt{g^2 - c} = 0$

$\Rightarrow \quad g = 0$

$x^2 + y^2 + 2fy = 0 \qquad \ldots (1)$

Differentiate w.r.t. x

$x + yy' + 2y'\left(\frac{-x^2 - xy^2}{2y}\right) = 0$ (by using (1))

$x + y'\left(y - \frac{-x^2 - y^2}{2y}\right) = 0$

$x + y'\left(\frac{y^2 - x^2}{2y}\right) = 0$

$y'(x^2 - y^2) = 2xy$

$\Rightarrow \quad g(x) = 2x$

3. **Ans (B)**

$|adj(adjA)| = |adjA|^2$

$= \left|2\sqrt{3}\right|^2 = 12$

4. **Ans (B)**

Required probability $= \frac{{}^3C_1}{{}^9C_1} \cdot \frac{{}^4C_1}{{}^8C_1} \cdot \frac{{}^2C_1}{{}^7C_1} \cdot 3! = \frac{2}{7}$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (A,B,D)**

Equation of normal is $y = mx - 2m - m^3$

It passes through the point (9, 6) then

$6 = 9m - 2m - m^3$

$\Rightarrow \quad m^3 - 7m + 6 = 0$

$\Rightarrow \quad (m - 1)(m - 2)(m + 3) = 0$

$\Rightarrow \quad m = 1, 2, -3$

Equations of normals are

$y - x + 3 = 0,$

$y + 3x - 33 = 0$

& $\quad y - 2x + 12 = 0$

6. **Ans (A,B,C)**

$a_{ij} = (i^2 + j^2 - ij)(j - i)$

$a_{ji} = (j^2 + i^2 - ji)(i - j) = -a_{ij}$

$\Rightarrow$ A is skew symmetric

$\Rightarrow T_r(A) = 0 \Rightarrow |A| = 0$

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7. **Ans (A,B,C,D)**

Given $\left|\frac{\alpha-\beta}{1-\bar{\alpha}\beta}\right|=1 \quad \left|\frac{\alpha-\beta}{1-\bar{\alpha}\beta}\right|^2=1^2$

$\left(\frac{\alpha-\beta}{1-\bar{\alpha}\beta}\right)\left(\overline{\frac{\alpha-\beta}{1-\bar{\alpha}\beta}}\right)=1$

$(\alpha-\beta)\left(\overline{\alpha-\beta}\right)=(1-\bar{\alpha}\beta)\left(\overline{1-\bar{\alpha}\beta}\right)$

$(\alpha-\beta)\left(\bar{\alpha}-\bar{\beta}\right)=(1-\bar{\alpha}\beta)\left(\bar{1}-\overline{\bar{\alpha}\beta}\right)$

$\alpha\bar{\alpha}-\alpha\bar{\beta}-\beta\bar{\alpha}+\beta\bar{\beta}=(1-\bar{\alpha}\beta)\left(1-\alpha\bar{\beta}\right)$

$|\alpha^2|-\alpha\bar{\beta}-\beta\bar{\alpha}+|\beta|^2=1-\alpha\bar{\beta}-\bar{\alpha}\beta+\alpha\bar{\alpha}\beta\bar{\beta}$

$|\alpha|^2+|\beta|^2=1+|\alpha|^2|\beta|^2$

$1-|\alpha|^2+|\alpha|^2|\beta|^2-(\beta)^2=0$

$(1-|\alpha|^2)-|\beta|^2(1-|\alpha|^2)=0$

$(1-|\alpha|^2)(1-|\beta|^2)=0$

$|\alpha|^2=1 \quad \text{or} \quad |\beta|^2=1$

$|\alpha|=1 \quad \text{or} \quad |\beta|=1$

$\alpha=e^{i\theta} \quad \text{or} \quad \beta=e^{i\theta}$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (156.50)**

Total natural number formed

$= 4 + 16 + 64 + 256 = 340$

Total natural number not exceeding 8531

= 4 (1 digit) + 16 (2 digit) + 64 (3 digit) + 229 (4 digit) = 313

2. **Ans (1.00)**

Total 4 digit (different) number = 24

number according to given condition

$= 4!\left(1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}\right)$

Required probabilty $=\frac{9}{24}=\frac{3}{8}=9$

3. **Ans (64.00)**

Equating of line L_1 and L_2 can be obtained in symmetric form

$L_1:\frac{x}{0}=\frac{y-b}{-b}=\frac{z}{c}$

$L_2:\frac{x-a}{a}=\frac{y}{0}=\frac{z}{c}$

Equation of plane p : $\begin{vmatrix} x & y-b & z \\ 0 & -b & c \\ a & 0 & c \end{vmatrix}=0$

$\Rightarrow \frac{x}{a}-\frac{y}{b}-\frac{z}{c}+1=0$

Lines L_1 and L_2 in vector form

$L_1:\bar{r}=b\hat{j}+\lambda(-b\hat{j}+c\hat{k})$

$L_2:\bar{r}=a\hat{j}+\mu(a\hat{i}+c\hat{k})$

Shortest Distance $=\frac{1}{4}=\frac{\begin{vmatrix} -a & b & 0 \\ 0 & -b & c \\ a & 0 & c \end{vmatrix}}{\sqrt{(bc)^2+(ac)^2+(ab)^2}}$

$\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=64$

Image of A(1, 0, 0) in the plane $x - y - z + 1 = 0$ is $\left(\frac{-1}{3},\frac{4}{3},\frac{4}{3}\right)$ its distance from M is 3.

4. **Ans (3.00)**

Equating of line L_1 and L_2 can be obtained in symmetric form

$L_1:\frac{x}{0}=\frac{y-b}{-b}=\frac{z}{c}$

$L_2:\frac{x-a}{a}=\frac{y}{0}=\frac{z}{c}$

Equation of plane p : $\begin{vmatrix} x & y-b & z \\ 0 & -b & c \\ a & 0 & c \end{vmatrix}=0$

$\Rightarrow \frac{x}{a}-\frac{y}{b}-\frac{z}{c}+1=0$

Lines L_1 and L_2 in vector form

$L_1:\bar{r}=b\hat{j}+\lambda(-b\hat{j}+c\hat{k})$

$L_2:\bar{r}=a\hat{j}+\mu(a\hat{i}+c\hat{k})$

Shortest Distance $=\frac{1}{4}=\frac{\begin{vmatrix} -a & b & 0 \\ 0 & -b & c \\ a & 0 & c \end{vmatrix}}{\sqrt{(bc)^2+(ac)^2+(ab)^2}}$

$\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=64$

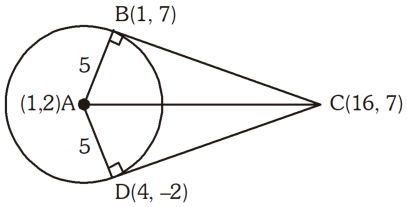
Image of A(1, 0, 0) in the plane $x - y - z + 1 = 0$ is $\left(\frac{-1}{3},\frac{4}{3},\frac{4}{3}\right)$ its distance from M is 3.

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PART-3 : MATHEMATICS

SECTION-II (ii)

5. Ans (75)



The eq. of circle is

$x^2 + y^2 - 2x - 4y - 20 = 0$

Centre (1, 2) radius = $\sqrt{1 + 4 + 20} = 5$

Using eq. of tangents at (x_1, y_1) of

$x^2 + y^2 + 2gx + 2fy + c = 0$ is

$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$

Eq. of tangent at (1, 7) is

$x \cdot 1 + y \cdot 7 - (x+1) - 2(y+7) - 20 = 0$

$\Rightarrow \quad y - 7 = 0$

Similarly eq. of tangent at (4, –2) is

$4x - 2y - (x + 4) - 2(y - 2) - 20 = 0$

$\Rightarrow \quad 3x - 4y - 20 = 0 \qquad .....(2)$

For pt C, solving (1) and (2), we get

$x = 16, y = 7$

$\therefore \quad C(16, 7)$

Now, clearly ar (quad ABCD) = 2 Ar (rt ΔABC)

$= 2 \times \frac{1}{2} \times AB \times BC = AB \times BC$

where AB = radius of circle = 5

and BC = length of tangent from C to circle

$= \sqrt{16^2 + 7^2 - 32 - 28 - 20} = \sqrt{225} = 15$

$\therefore \quad$ ar (quad ABCD) = 5 × 15 = 75 sq. units

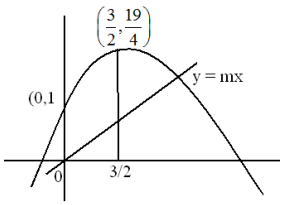
6. Ans (26)

$$\text{Total area} = \int_0^{3/2} (1 + 4x - x^2)\,dx$$

$$= x + 2x^2 - \frac{x^3}{3}\Big|_0^{3/2} = \frac{39}{8}$$

$$\& \ \frac{39}{16} = \frac{1}{2} \cdot \frac{3}{2} \cdot \frac{3}{2} m$$

$$\Rightarrow 3m = \frac{13}{2} \Rightarrow 12m = 26$$



7. Ans (2)

$(\tan y + y\sec^2 y)dy = e^x(\sin x + \cos x)dx$

$y\tan y = e^x \sin x + C$

(0,0) gives $C = 0$

$y\tan y = e^x \sin x$

$\pi \tan \pi = e^{x_1} \sin x_1$

$\Rightarrow \sin x_1 = 0 \ \Rightarrow \ x_1 = 2\pi$

$\frac{x_1}{\pi} = 2$

8. Ans (2)

The chord joining α and β is

$$\frac{x\cos[(\alpha+\beta)/2]}{a} + \frac{y\sin[(\alpha+\beta)/2]}{b}$$

$$= \cos\left(\frac{\alpha-\beta}{2}\right)$$

$$= \cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

If $\frac{\alpha+\beta}{2} = \theta$

then the equation of the chord is

$$\frac{x}{(a/\sqrt{25})}\cos\theta + \sin\theta = 1$$

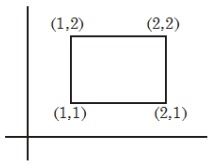
Hence, the chord touches the ellipse

$$\frac{x^2}{(a/\sqrt{2})^2} + \frac{y^2}{(b/\sqrt{2})^2} = 1$$

or $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{1}{2}$

Therefore, k = 2

9. Ans (1)



10. Ans (81)

$$\begin{vmatrix} x-1 & y-0 & z-1 \\ 1-1 & 2 & 1-1 \\ 1-0 & 0-1 & 1+2 \end{vmatrix} = 0$$

$\Rightarrow \quad 3x - z - 2 = 0$

$\vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma k$ || to $3x - z - 2 = 0$

$\Rightarrow \quad 3\alpha - 8 = 0 \qquad .....(1)$

$\vec{a} \perp \hat{i} + 2\hat{j} + 3k$

$\Rightarrow \quad \alpha + 2\beta + 38 = 0 \qquad ..... (2)$

$\vec{a}\left(\hat{i} + \hat{j} + 2k\right) = 0$

$\Rightarrow \quad \alpha + \beta + 28 = 2 \qquad ..... (3)$

on solving 1, 2 & 3

$\alpha = 1, \quad \beta = -5, \quad 8 = 3$

So $(\alpha - \beta + 8) =$

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